

AUTOMATIC LPG GAS LEAKAGE Detection And Protection

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LPG (liquid petroleum gas) used in kitchen gas cylinders is a mixture of propane and butane in diverse composition. Even though LPG is important for use

in kitchens at home/hotels/restaurants, and the public is usually alert while using it, but still some steps need to be introduced for safety. LPG used in cylinders (propane or

butane) is highly combustible and a danger whenever it leaks from faulty equipment such as gas regulators, gas pipes/tubes, and other components of LPG gadgets/devices. The gas tube that fetches gas from the cylinder to the burner is generally the first to malfunction as it experiences the most damage/erosion at



Fig. 1: Author's prototype

BILL OF MATERIAL

Components	Quantity
Arduino Uno (MOD1)	1
12V, 2A adaptor	1
SIM900A GSM module (MOD2)	1
IC ULN2003 (IC1)	1
5V relay (R1)	1
Buzzer (B1)	1
MQ6 or MQ2 sensor (S2)	1
MG996R high torque servo motor (S1)	1
Breadboard	1

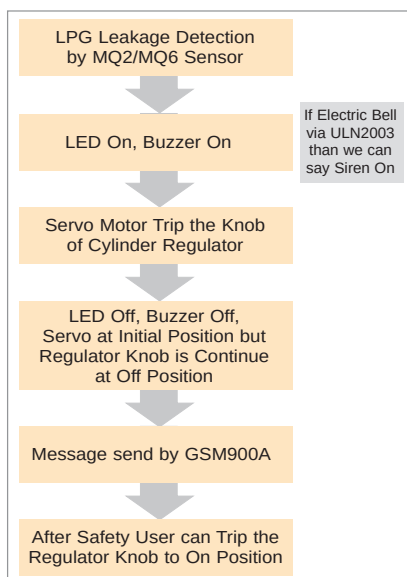


Fig. 2: Flow chart of the project

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IPG_GAS.ino
1 //RAKESH JAIN PROGRAM TO DETECT AND PROTECT FROM LPG/CNG GAS LEAKAGE
2 #include <Servo.h>;
3 #include <SoftwareSerial.h>;
4 SoftwareSerial mySerial (7, 8);
5 Servo myservo;
6 int pos = 20;
7 int LED = 12;
8 int BUZZER = 13;
9 int LPG_sensor = 3; // MQ-6 SENSOR
10 int LPG_detected;
11 char msg;
12 void setup()
13 {
14   mySerial.begin (9600) ;
15   Serial.begin (9600) ; // start serial debugging
16   myservo.attach(11);
17   Serial.begin(9600);
18   pinMode(LED, OUTPUT);
19   pinMode(BUZZER, OUTPUT);
20   pinMode(LPG_sensor, INPUT);
21   myservo.write(pos);
22 }
  
```

Fig. 3: Code snippet defining the pins